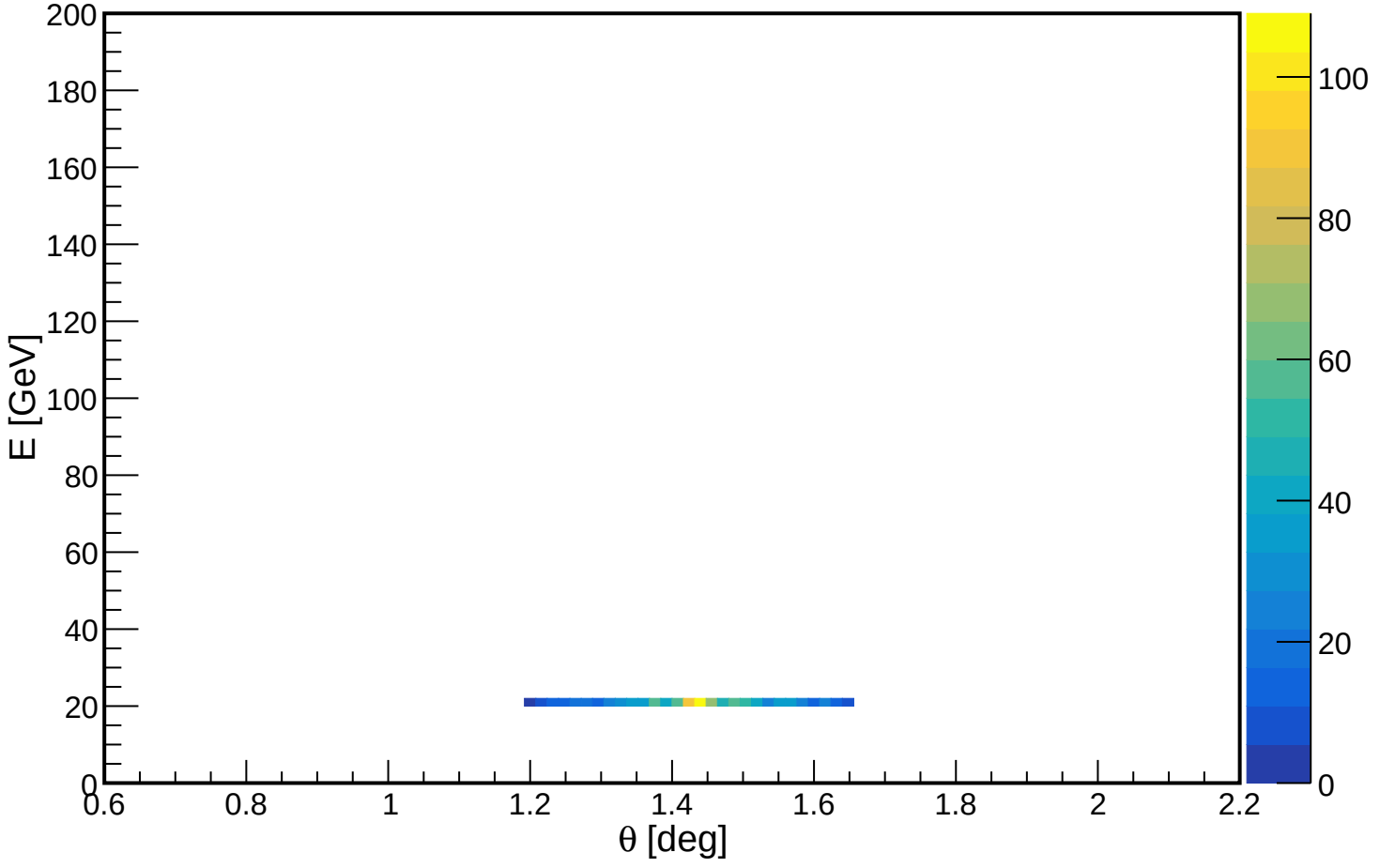
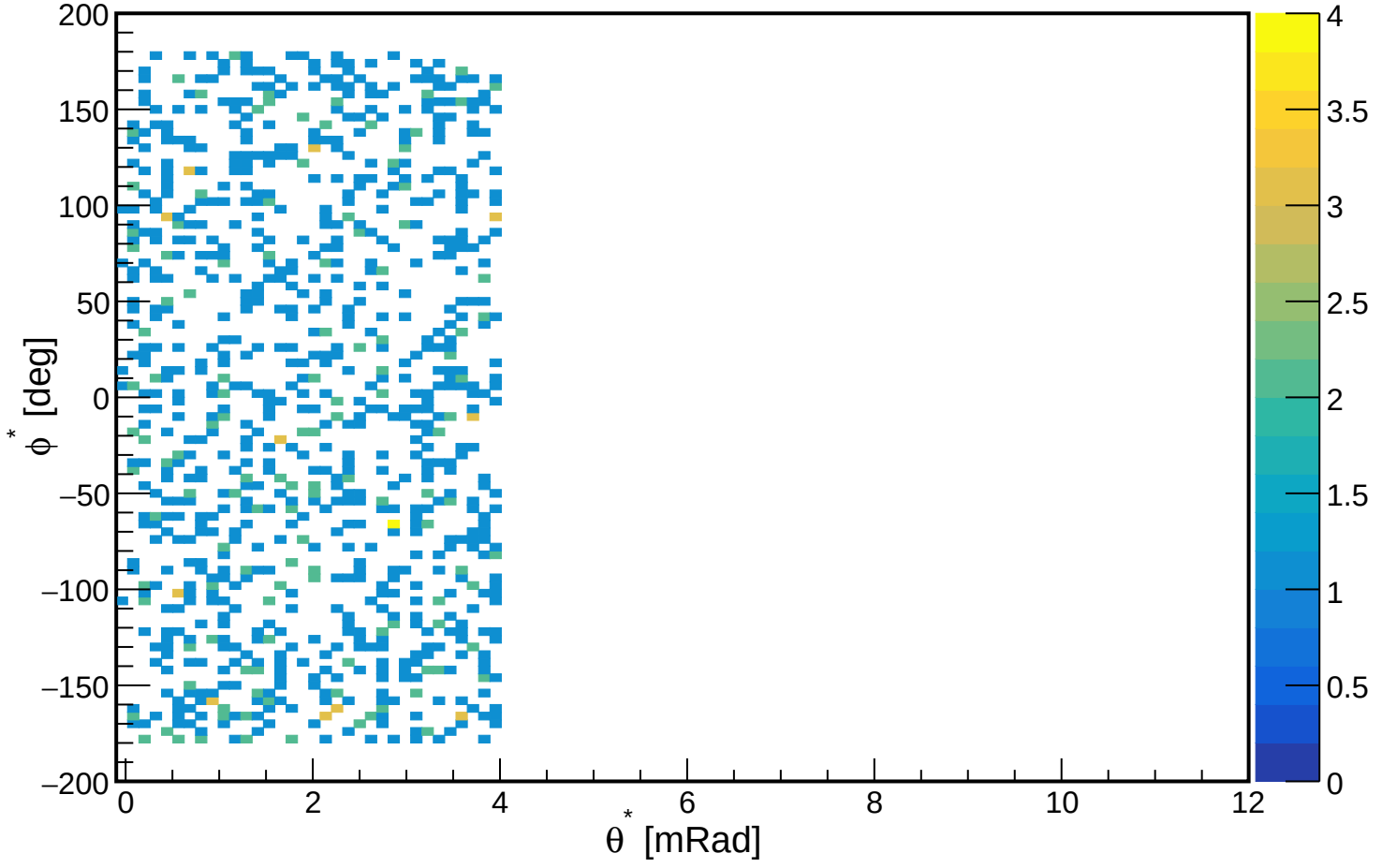
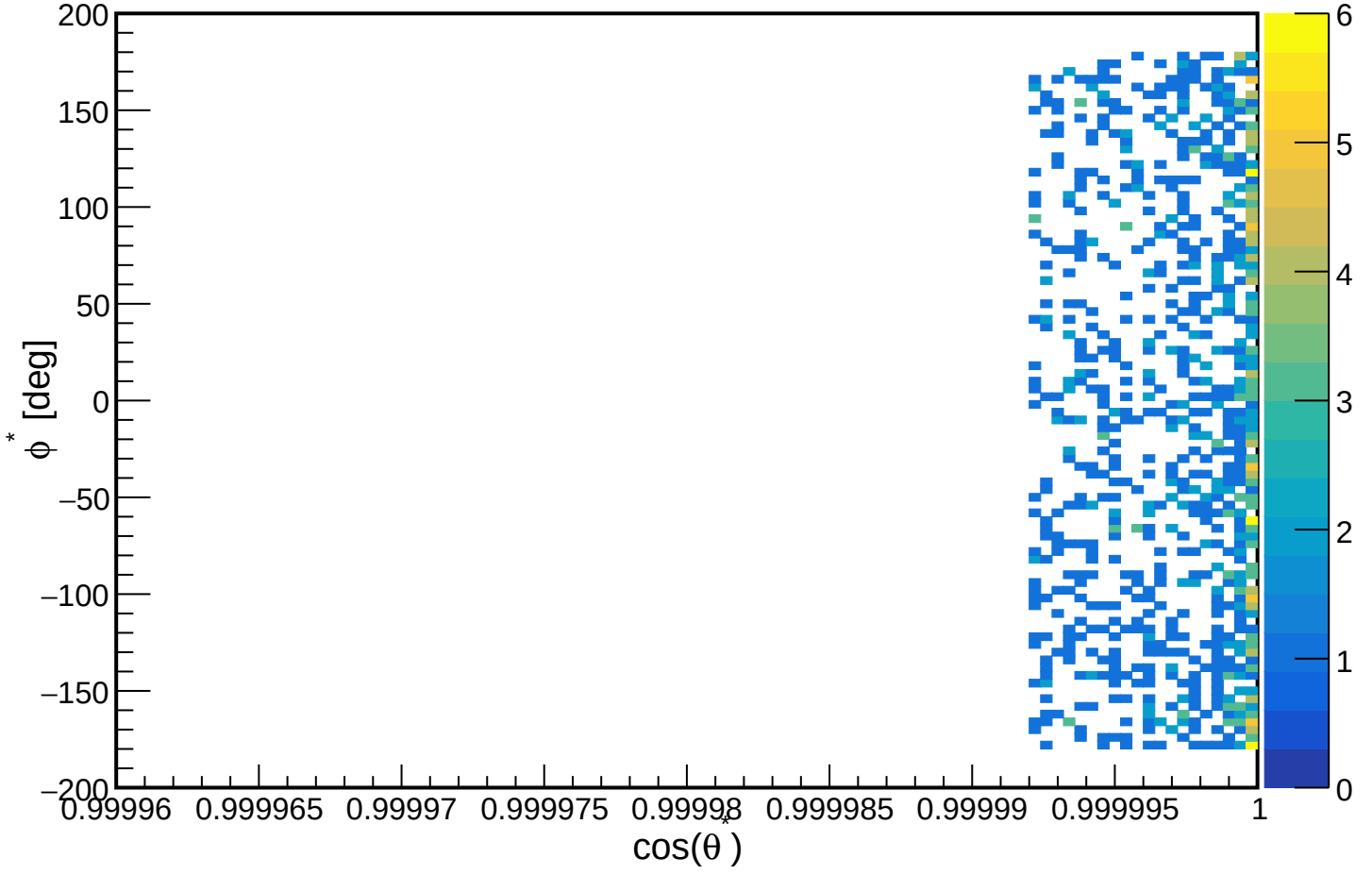
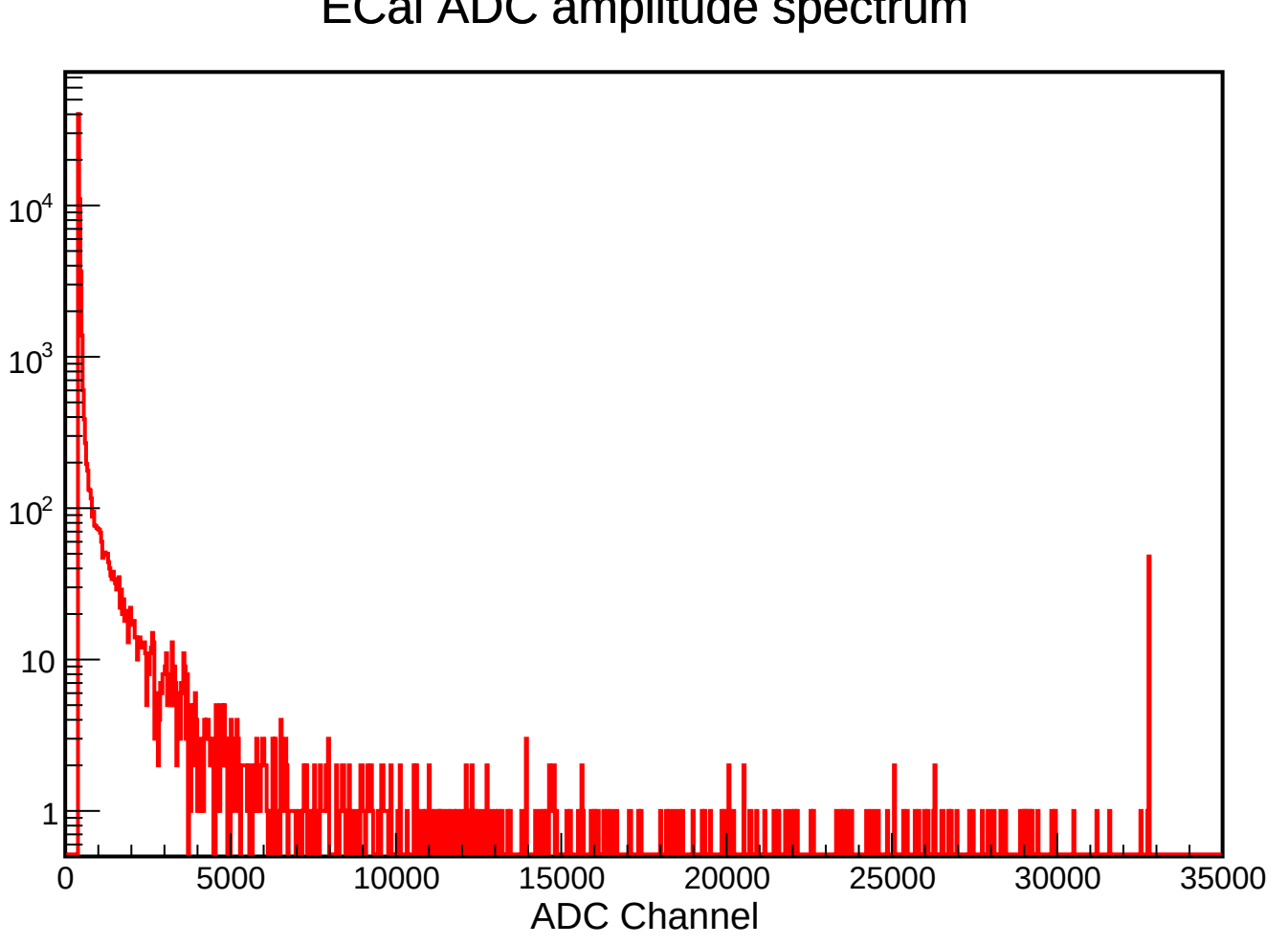
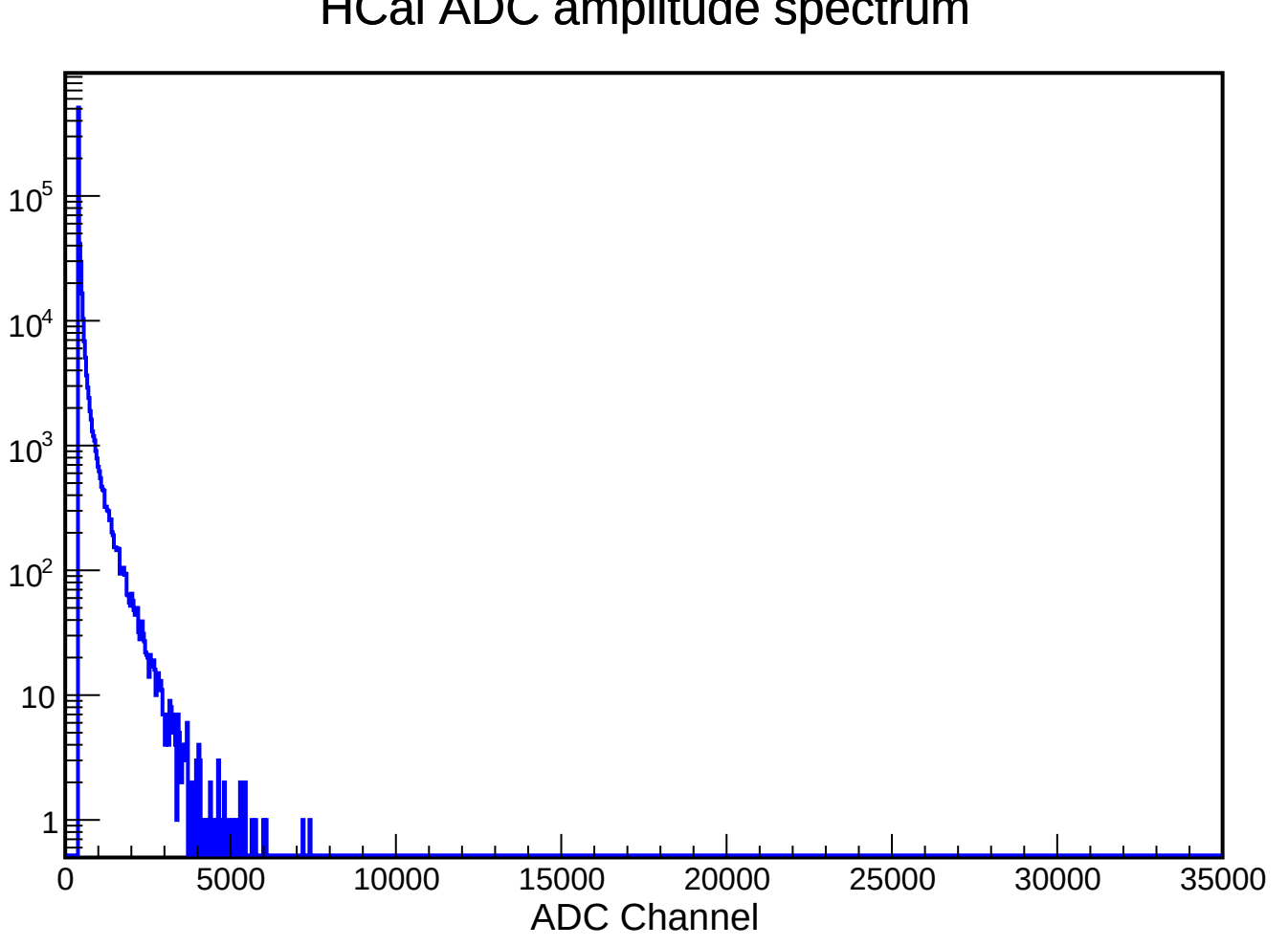
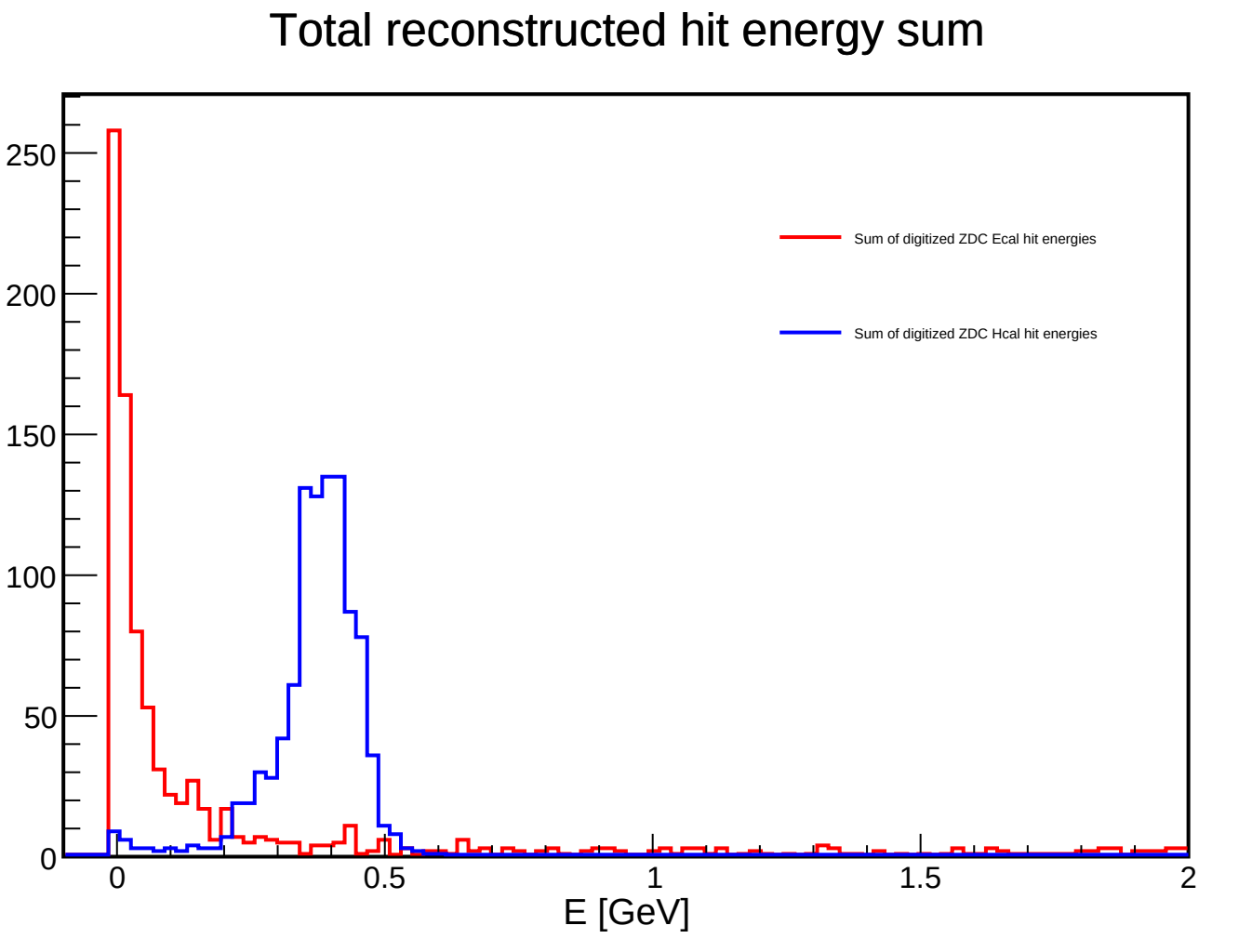

Neutron true energy vs. polar angle
E [GeV]
θ [deg]
0
20
40
60
80
100
120
140
160
180
200
0.6
0.8
1
1.2
1.4
1.6
1.8
2
2.2
0
20
40
60
80
100


Neutron true azimuthal angle vs. polar angle around p axis
200
150
100
50
0
−50
−100
−150
−200
$\phi^*$ [deg]
0
2
4
6
8
10
12
$\theta^*$ [mRad]
4
3.5
3
2.5
2
1.5
1
0.5
0

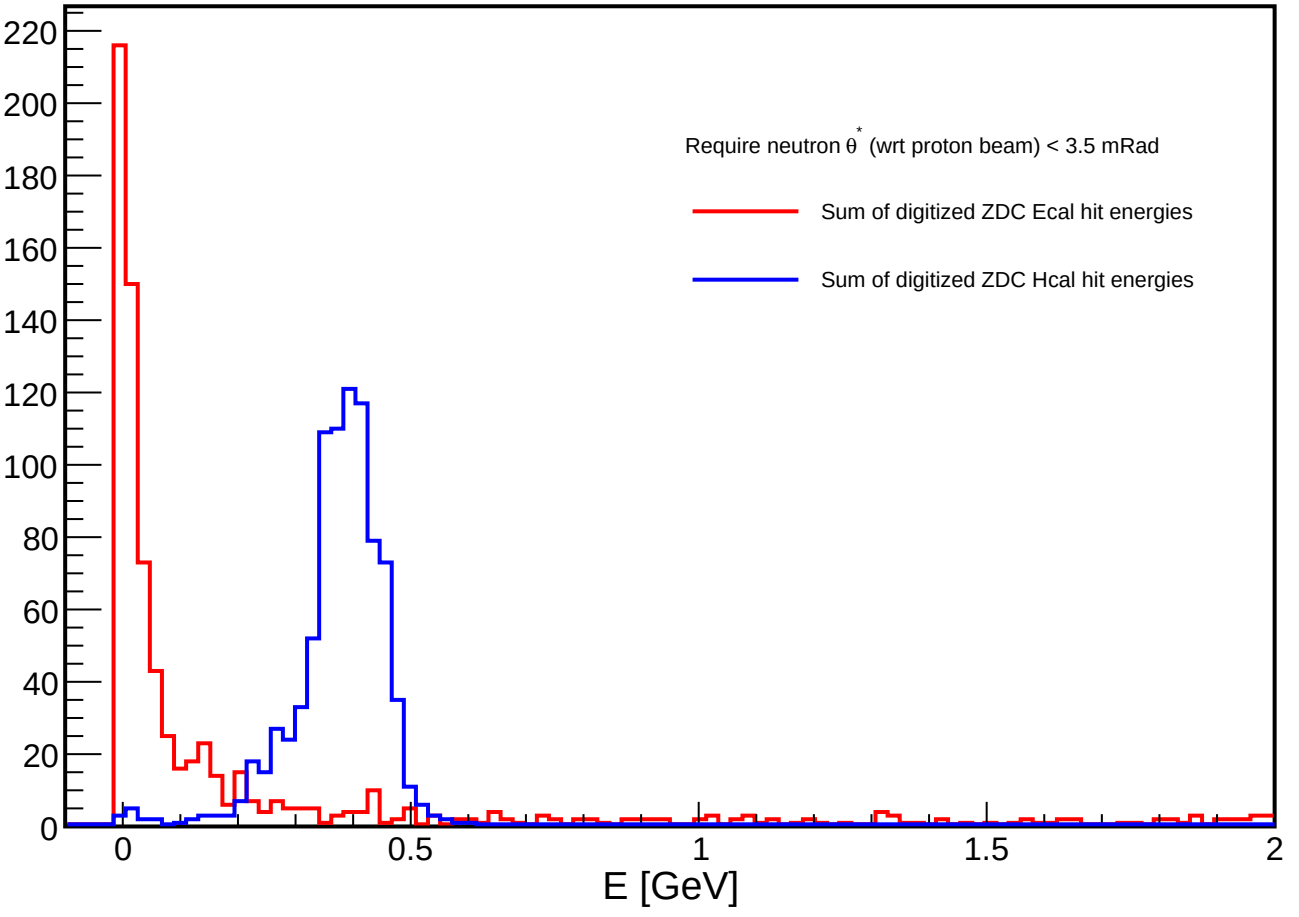

Neutron true azimuthal angle vs. cosine of polar angle around p axis
200
150
100
50
0
−50
−100
−150
−200
$\phi^*$ [deg]
0.99996
0.999965
0.99997
0.999975
0.99998
0.999985
0.99999
0.999995
1
$\cos(\theta^*)$
6
5
4
3
2
1
0

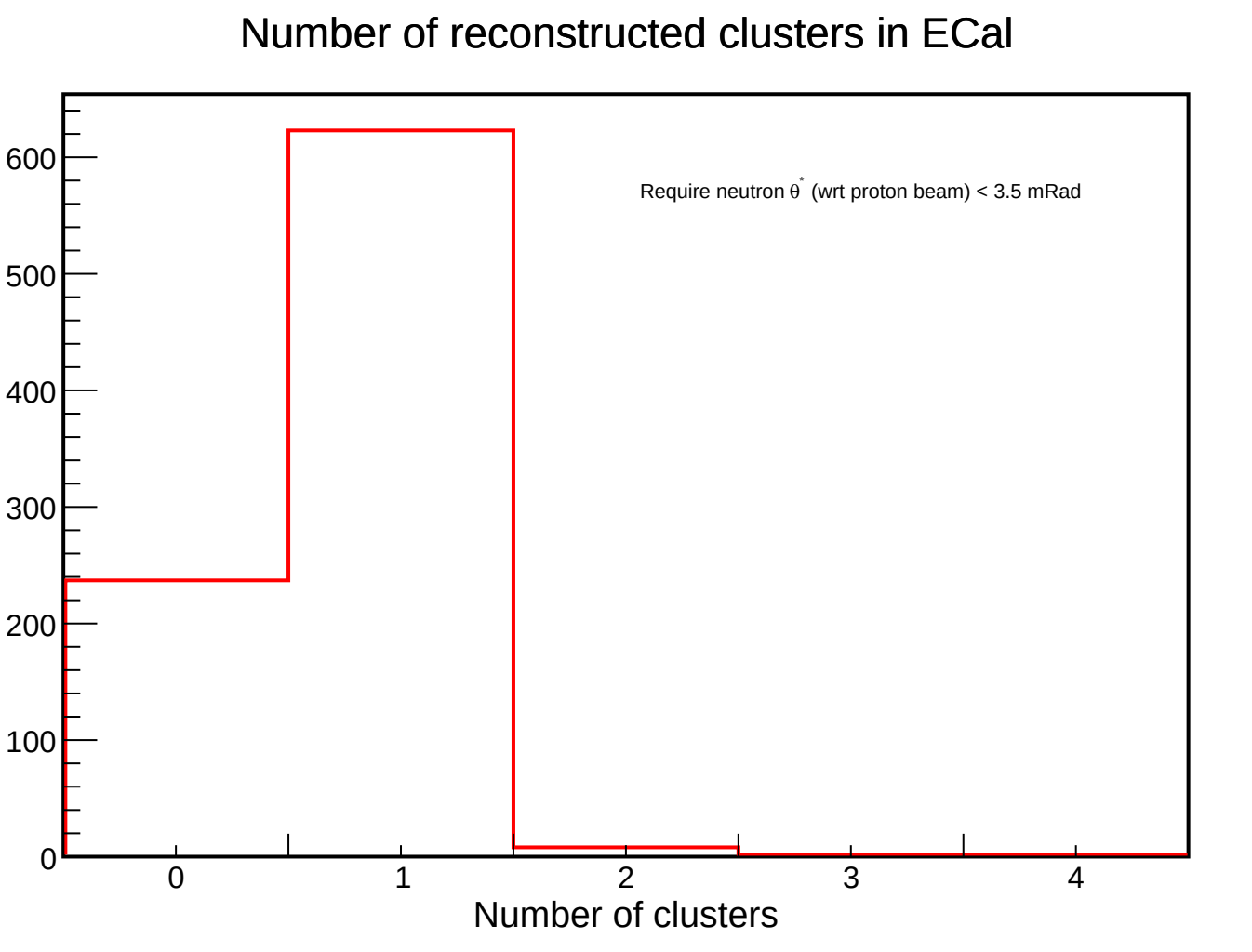

ECal ADC amplitude spectrum
$10^4$
$10^3$
$10^2$
10
1
0
5000
10000
15000
20000
25000
30000
35000
ADC Channel

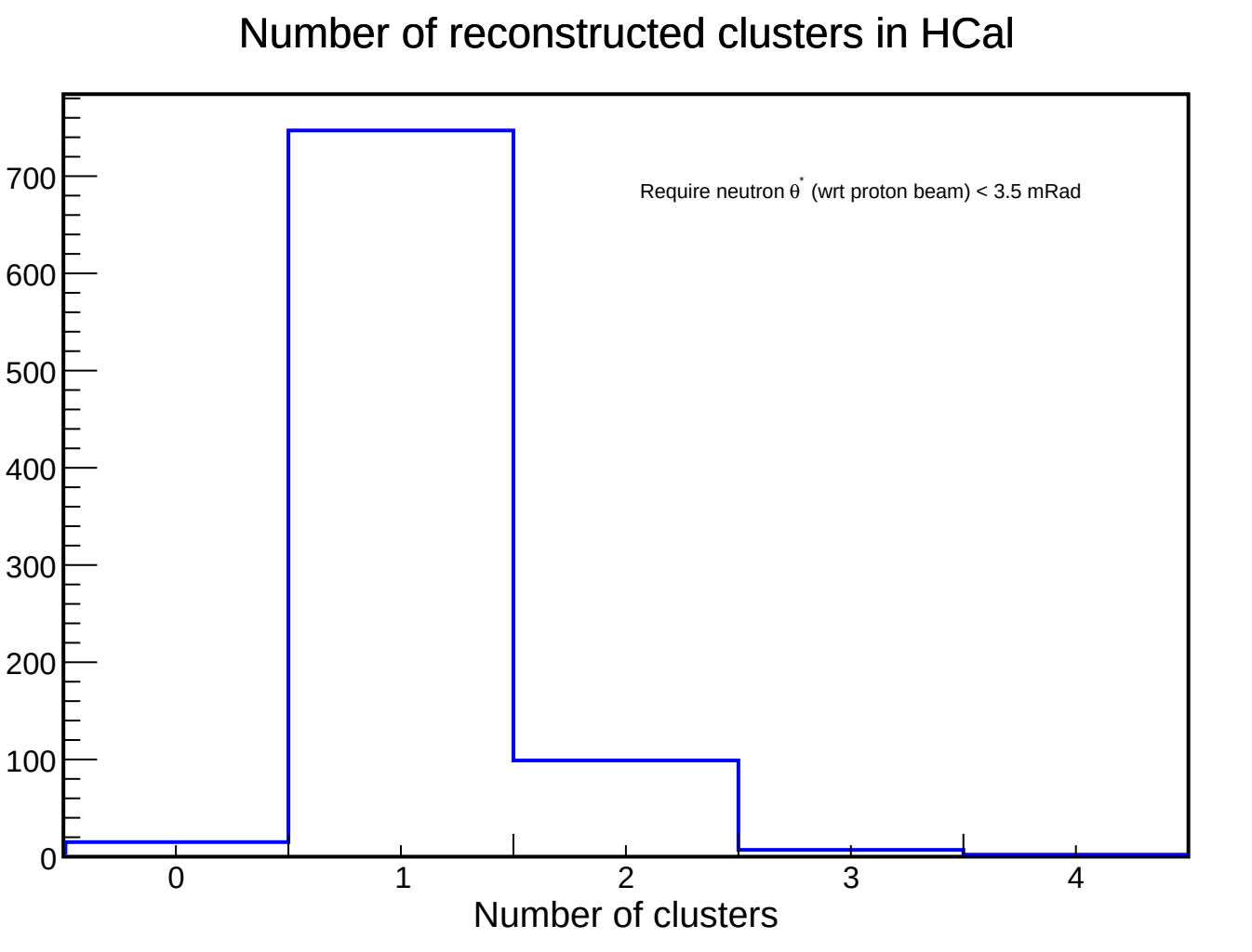

HCal ADC amplitude spectrum
$10^5$
$10^4$
$10^3$
$10^2$
10
1
0
5000
10000
15000
20000
25000
30000
35000
ADC Channel

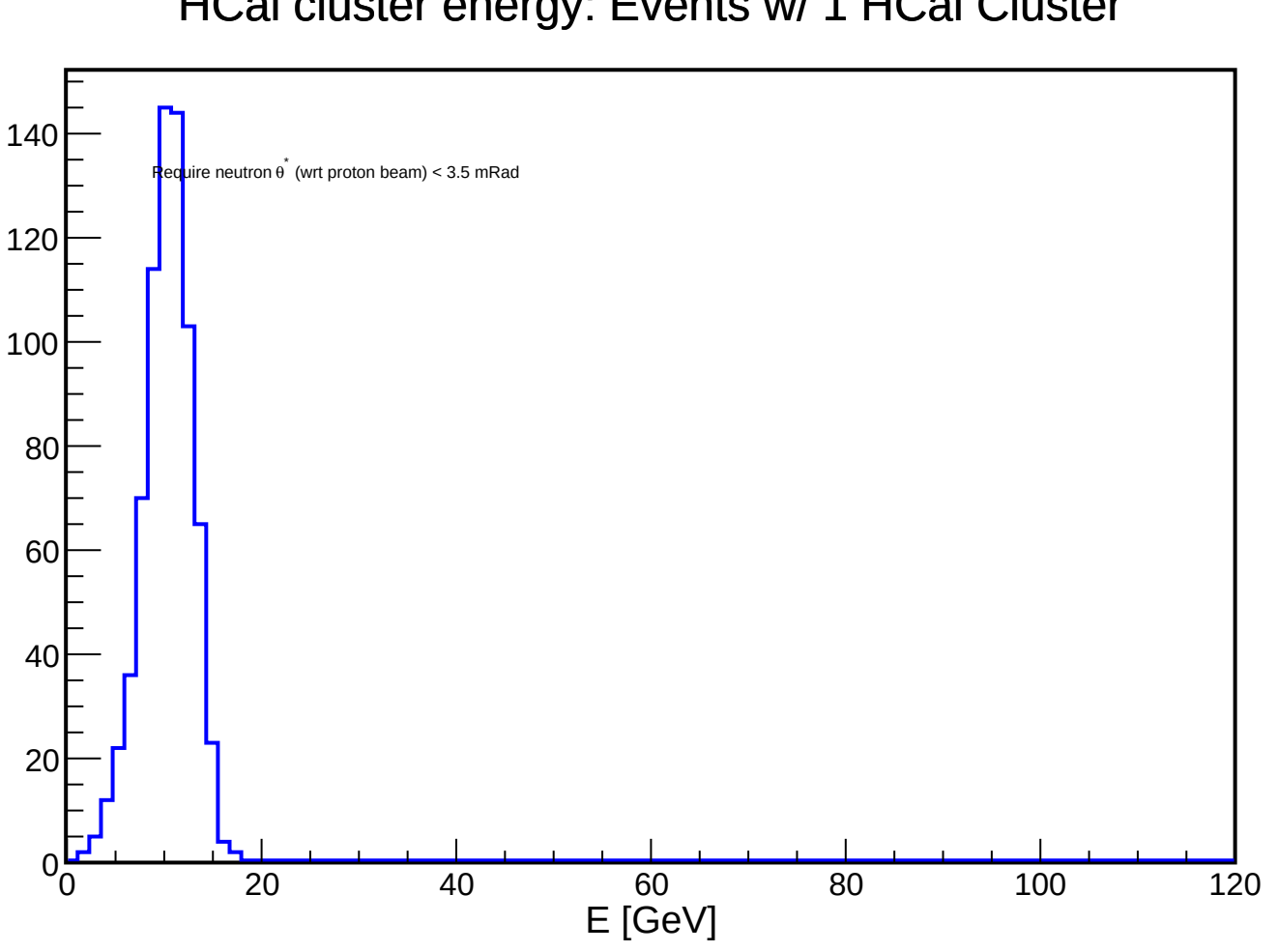

Total reconstructed hit energy sum
Sum of digitized ZDC Ecal hit energies
Sum of digitized ZDC Hcal hit energies
250
200
150
100
50
0
0
0.5
1
1.5
2
E [GeV]

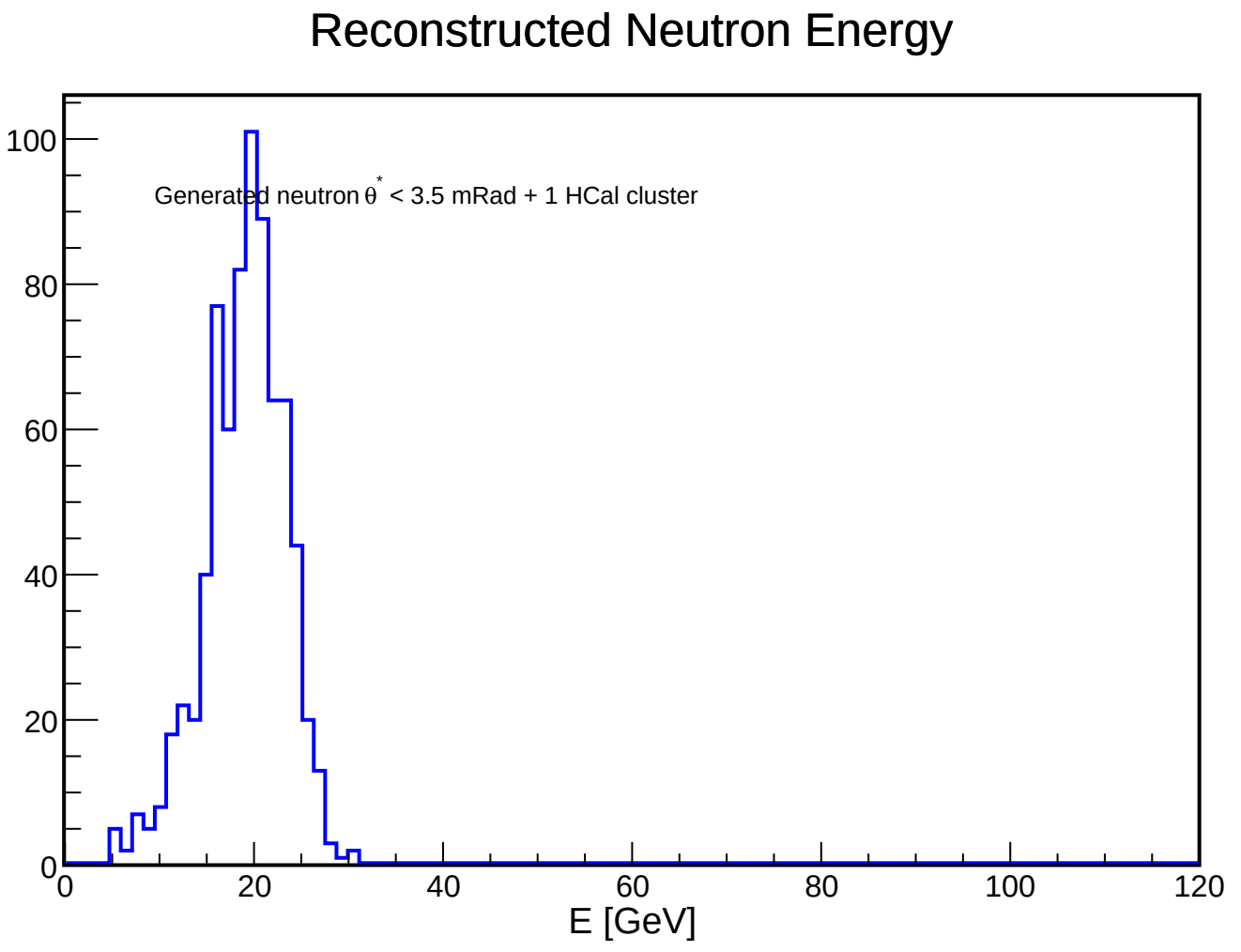
# Total reconstructed hit energy sum

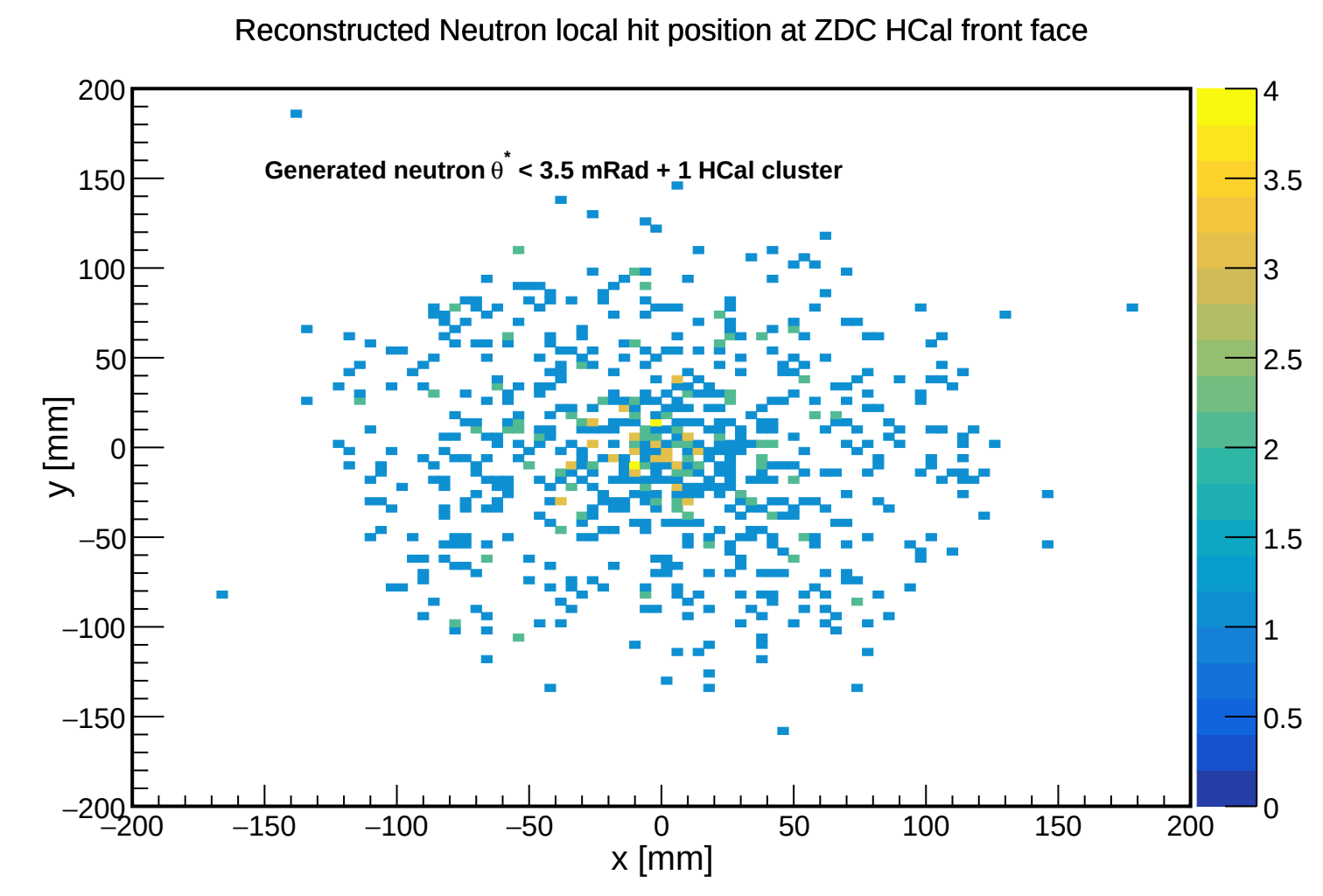



Number of reconstructed clusters in ECal
Require neutron θ* (wrt proton beam) < 3.5 mRad
600
500
400
300
200
100
0
0
1
2
3
4
Number of clusters


Number of reconstructed clusters in HCal
Require neutron θ* (wrt proton beam) < 3.5 mRad
700
600
500
400
300
200
100
0
0
1
2
3
4
Number of clusters


HCal cluster energy: Events w/ 1 HCal Cluster
Require neutron θ* (wrt proton beam) < 3.5 mRad
140
120
100
80
60
40
20
0
0
20
40
60
80
100
120
E [GeV]


Reconstructed Neutron Energy
Generated neutron θ* < 3.5 mRad + 1 HCal cluster
100
80
60
40
20
0
0
20
40
60
80
100
120
E [GeV]


Reconstructed Neutron local hit position at ZDC HCal front face
Generated neutron θ* < 3.5 mRad + 1 HCal cluster
y [mm]
x [mm]
200
150
100
50
0
−50
−100
−150
−200
4
3.5
3
2.5
2
1.5
1
0.5
0